on the file you want the DSP to run, and the EZ-KIT Lite host takes care of the rest, downloading the program and starting it running.

This brings us to two questions. First, how do we test our audio filter to make sure it is operating as we designed it; and second, what in the world is a company called *Analog Devices* doing making *Digital Signal Processors*?

Analog measurements on a DSP system

For just a few moments, forget that you are studying *digital* techniques. Let's take a look at this from the standpoint of an engineer that specializes in *analog* electronics. He doesn't care what is inside of the EZ-KIT Lite, only that it has an analog input and an analog output. As shown in Fig. 29-3, he would invoke the traditional analog method of analyzing a "black box," attach a signal generator to the input, and look at the output on an oscilloscope.

What does our analog guru find? First, the system is *linear* (as least as far as this simple test can tell). If a sine wave is placed into the input, a sine wave is observed on the output. If the amplitude or frequency of the input is changed, a corresponding change is seen in the output. When the input frequency is slowly increased, there comes a point where the amplitude of the output sine wave decreases rapidly to zero. That occurs just below one-half the sampling rate, due to the action of the anti-alias filter on the ADC.

Now our engineer notices something unknown in the analog world: the system has a perfect *linear phase*. In other words, there is a constant delay between an event occurring in the input signal, and the result of that event in the output signal. For instance, consider our example filter kernel in Fig. 29-3. Since the center of symmetry is at sample 150, the output signal will be delayed by 150 samples relative to the input signal. If the system is sampling at 8 kHz, for example, this delay will be 18.75 milliseconds. In addition, the sigma-delta converter will also provide a small additional fixed delay.

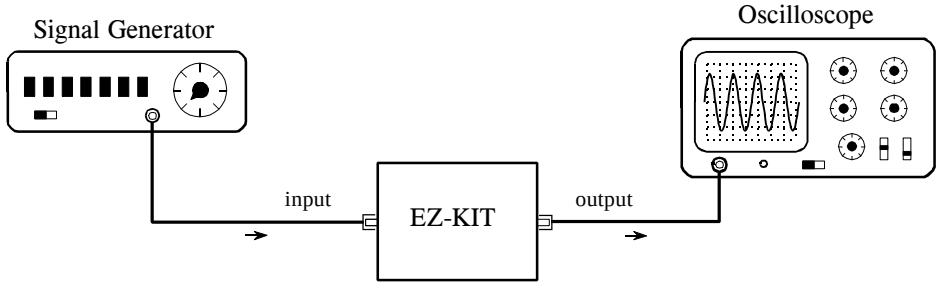

FIGURE 29-3
Testing the EZ-KIT Lite. Analog engineers test the performance of a system by connecting a signal generator to its input, and an oscilloscope to its output. When a DSP system (such as the EZ-KIT Lite) is tested in this way, it appears to be a virtually perfect analog system